

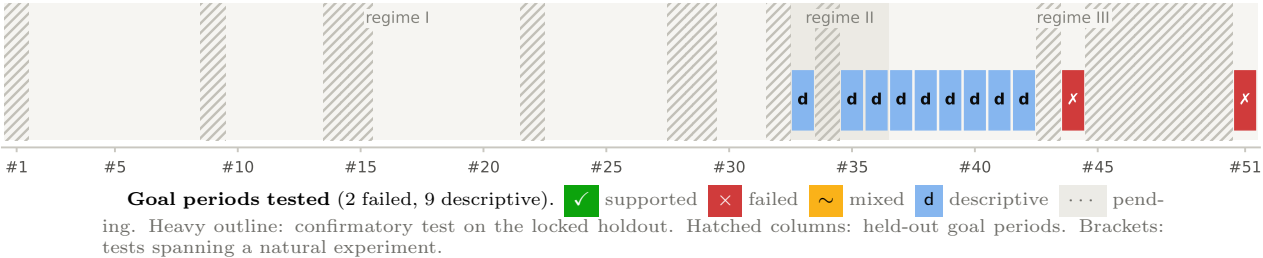
# H66: Shared platform latency is a common-noise field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

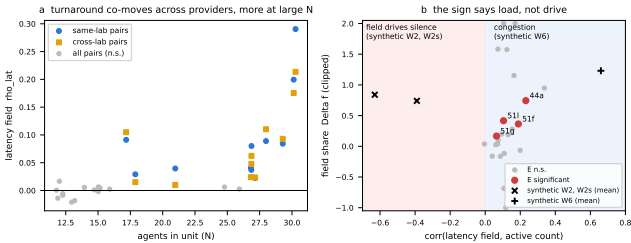
**Bottom line.** The residual co-activation after trimming is small; the platform’s common latency field is real but follows activity rather than driving it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

After the day is trimmed to the window in which every agent runs, little co-activation is left in regime III. The question is whether a common platform-latency field explains that residual. The test uses call turnaround from the DQ1 ledger on 29 non-holdout units (regimes II and III) and Gemini’s logged server time. The residual beats the trimmed block-shift null in 4/29 units. A latency field exists in 11/29 units, all in #44 and #51, and it grows with  $N$ . It is not a provider field: same-lab and cross-lab pairs co-move alike. It is not a driver either. It rises when more agents are active (28/29 units), which is the opposite of the model’s sign. The pre-registered rule says mixed; a post hoc sign test and a congestion world say the field share measures load. **Refuted as posed.** The holdout is not run.



**Observables.** (a) Cross-agent turnaround correlation per regime-III unit vs  $N$ . Same-lab and cross-lab pairs co-move alike, so the field is not a provider’s API. (b) Field share  $\Delta f$  against the sign of the field’s relation to the active count. A field that silences agents gives a negative sign (synthetic W2). Congestion gives a positive sign and  $\Delta f \approx 1$  (W6). Every real unit with a residual sits on the load side.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not blame platform slowness for agents acting together. On the call clock, the platform’s latency field is a load meter: it rises when more agents run, and it does not make them co-act. A field share from a third-party-field regression is uninformative without the sign of the field–activity relation; report both. For a scaffold builder, the useful number is the growth of turnaround co-movement with  $N$ : about 0 at  $N \leq 16$ , 0.02–0.10 at  $N$  17–28, and 0.2 at  $N = 30$ . Watch it as a capacity signal. Gemini’s API time is only 0.44 of a call’s turnaround, and it does not co-move across agents; the shared part is the village’s own execution layer.

## PHYSICAL MODEL

Models 01 and 02. The spin  $a_i(m)$  is 1 when agent  $i$  logs an act call in minute  $m$  of the all-present window. The latency spin  $z_i(m)$  is the agent-day-standardized log turnaround of its chained calls. The field  $L_{-ij}$  is the mean of  $z_k$  over third parties  $k \neq i, j$ , with lags 0–5 min, plus an infrastructure-error count  $u$ .  $h_i(b)$  is a 30-min block field.  $E$  is the mean pair correlation of block-demeaned spins above the block-shift null.  $K$  is the number of active agents.

$$P(\mathbf{a}) \propto \exp \left[ \sum_i (h_i(b) + \sum_{k=0}^5 \gamma_{ik} L(m-k) + \kappa_i u(m)) a_i + \frac{J_0}{2N} (\sum_i a_i)^2 \right]$$
$$f_{\text{lat}} = 1 - E_{\text{adj}}/E, \quad \Delta f = f_{\text{lat}} - \langle f_{\text{lat}} \rangle_{\text{shifted field}}$$

field drives silence:  $\text{corr}(L, K) < 0$ ; load:  $\text{corr}(L, K) > 0$

H66-platform-latency-field · generated 2026-10-04 13:01 by infra/summaries/build\_summaries.py · card hypotheses/H66-platform-latency-field/README.md · ratings: H66 agent (suggested)

## RESULTS VS PREDICTIONS

| Prediction (dated)                           | Observed                                                                          | Verdict       |
|----------------------------------------------|-----------------------------------------------------------------------------------|---------------|
| P0 synthetic recovery                        | sizes 0/16, 1/16; $\Delta f$ a lower bound (W2 0.74 vs 1.0); no false attribution | partly        |
| P1 field in $\geq 2/3$ units; provider field | 11/26 units; same-lab 0.080 vs cross-lab 0.062                                    | failed        |
| P2 $\Delta f < 0.25$ where $E$ sig.          | 4/29 sig.; $\Delta f$ 0.74, 0.36, 0.17, load (A2) 0.42; rule: mixed               | failed        |
| Post hoc sign test                           | $\text{corr}(L, K) > 0$ in 28/29 (median +0.11); W2 gives $-0.34$ to $-0.66$      | against H66   |
| P3 zero-lag field peak                       | $C_\ell(0)$ 0.023 vs $C_\ell(\pm 1)$ 0.017                                        | held          |
| P4 intensity $>$ binary                      | 14/29                                                                             | failed        |
| P5 same-room $>$ cross-room $E$              | 5/18 two-room units                                                               | failed        |
| P6 residual at window edges                  | 3/4 units ( $\times 2.0$ –6.1)                                                    | held          |
| G51 server time (native)                     | $r$ 0.49 with turnaround; no cross-agent co-movement                              | failed as H66 |
| G44 separate-stack leader                    | loading $-0.11$ vs median 0.14 (70 min)                                           | descriptive   |

## GOAL PERIODS IN DETAIL

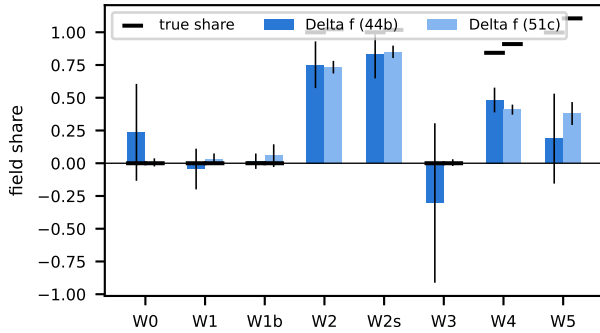
G33 d descriptive  
G35 d descriptive  
G36 d descriptive  
G37 d descriptive  
G38 d descriptive  
G39 d descriptive  
G40 d descriptive  
G41 d descriptive  
G42 d descriptive  
G44  $\times$  failed

+1 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



**Second figure.** Synthetic swarms on real unit skeletons (44b, 51c; 8 seeds). Bars:  $\Delta f$ ; ticks: true field share. The estimator stays near 0 when the field does not act (W1, W3). It recovers 0.4–0.85 of a field that drives pauses (W2, W2s, W4).

## CAVEATS

The sign test and the congestion world are post hoc (Amendment 2); they reverse a pre-registered “mixed”. Turnaround mixes generation, tool execution and overhead; only Gemini logs server time. Load may be congestion or shared heavy work; the data do not separate them. Only 4 units carry a residual, and their  $\Delta f$  intervals are wide. The separately served leader has 70 minutes. The holdout is not run.

## ROUND 2

Fit a load curve (latency vs active count) per unit and its slope vs  $N$ . Use provider-specific error bursts as exogenous latency shocks. Run `confirm.py` after sign-off.

- **Essence.** separating exogenous platform fields from swarm-made fields in the residual co-activation.
- **H66-R1** Congestion as an order parameter.
- **H66-R2** An exogenous platform probe.
- **H66-R3** Separate execution from generation with the Anthropic and OpenAI token counts (output tokens per second) where usage is logged.
- **H66-R4** Holdout.

H66-platform-latency-field · generated 2026-10-04 13:01 by infra/summaries/build\_summaries.py · card hypotheses/H66-platform-latency-field/README.md · ratings: H66 agent (suggested) (page 2)